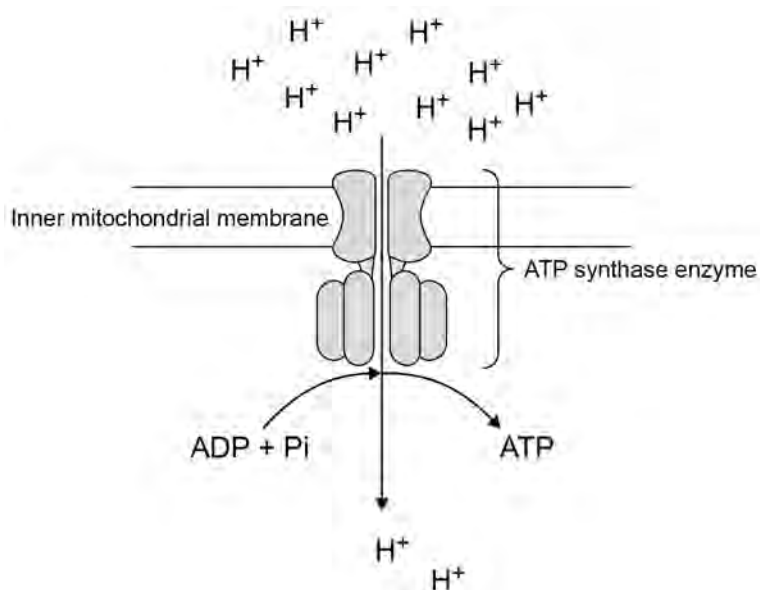


0 2 . 1 **Figure 1** shows an ATP synthase enzyme in the inner mitochondrial membrane.

Figure 1



Complete the passage with the appropriate terms.

[2 marks]

ATP synthase comprises several polypeptides, so is said to have

a _____ structure.

It catalyses the synthesis of an ATP molecule by a _____

reaction; this involves the _____ of a water molecule.

The ATP synthase in **Figure 1** is in a mitochondrion so would catalyse reactions during _____.



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As shown in **Figure 1**, ATP synthase has two functions.

- It catalyses the synthesis of ATP.
- It allows the movement of H^+ ions.

Suggest how the shape of the ATP synthase allows it to have these two functions.

Explain your answers.

[4 marks]

Catalyses the synthesis of ATP _____

Allows the movement of H^+ ions _____

6

Turn over for the next question

Turn over ►

